

1. Prerequisites:

Source Database Configuration:

Enable Binary Logging:

- log_bin = ON
- binlog_format = ROW
- binlog_row_image = FULL
- server_id configured
- User Permissions

Migration user requires:

- SELECT
- REPLICATION CLIENT
- REPLICATION SLAVE

Network Requirements:

- DMS can access EC2 MySQL.
- DMS can access RDS MySQL.
- Security Groups allow port 3306.

2. Migration Strategy:

Phase 1: Environment Preparation

Activities:

- Provision Amazon RDS MySQL.
- Create DMS Replication Instance.
- Configure Security Groups.
- Configure Route Tables if required.
- Create DMS Endpoints.
- Verify connectivity.

Deliverable:

Environment readiness sign-off.

Phase 2: Full Load Migration

Activities:

- Create a DMS Migration Task.
- Select Full Load + CDC mode.
- Start migration.
- Monitor progress.

Expected Outcome:

- All existing tables copied to RDS.

Deliverable:

Full Load Completion Report.

Phase 3: CDC Synchronization

Activities:

- Enable ongoing replication.
- Capture Inserts.
- Capture Updates.
- Capture Deletes.

Expected Outcome:

- Target remains synchronized with source.

Deliverable:

CDC Validation Report.

Phase 4: Data Validation

Validation Activities:

- Record Count Validation: Compare row counts between source and target.
- Sample Data Validation: Validate critical tables.
- Business Validation: Validate:
 - Customer Data
 - Account Data
 - Transaction Data
 - Reference Tables

Deliverable:

Migration Validation Report.

Phase 5: Cutover

Pre-Cutover

- Inform stakeholders.
- Schedule maintenance window.
- Take database backup.
- Verify CDC latency.

Cutover Steps

1. Stop application writes.
2. Allow CDC to catch up.
3. Confirm replication latency = 0.
4. Stop the DMS task.

5. Update application connection string.
6. Redirect application to RDS.
7. Perform smoke testing.

Deliverable:
Production Cutover Sign-Off.

3. Rollback Plan:

Rollback will be initiated if:

- Critical validation failures occur.
- Application connectivity issues persist.
- Data inconsistency is detected.

Rollback Steps:

1. Stop application.
2. Revert connection string to EC2 MySQL.
3. Restart application.
4. Notify stakeholders.
5. Investigate migration issues.

Expected Recovery Time:
Less than 30 minutes.

4. Risk Assessment:

Risk	Impact	Mitigation
Network Connectivity Failure	High	Pre-migration connectivity testing
DMS Replication Failure	High	Continuous monitoring
Data Mismatch	High	Validation procedures
Application Connection Issues	Medium	Pre-cutover testing
Performance Degradation	Medium	Performance benchmarking

5. Monitoring Plan:

AWS DMS Metrics:

Monitor:

- Full Load Progress

- CDC Source Latency
- CDC Target Latency
- Task Status
- Replication Errors
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Database Monitoring

Monitor:

- CPU Utilization
- Memory Usage
- Connections
- Storage Usage
- Slow Queries

6. Success Criteria:

Migration will be considered successful when:

- All databases migrated.
- All tables migrated.
- Data validation completed successfully.
- Application functions correctly on RDS.
- No critical defects identified.
- Stakeholder approval obtained.

7. Migration Timeline:

Phase	Duration
Environment Setup	1 Day
DMS Configuration	0.5 Day
Full Load Migration	Depends on Data Volume
CDC Synchronization	1–7 Days
Validation	1 Day
Cutover	1 Hour
Post-Migration Support	3 Days

8. Deliverables:

1. Migration Architecture Document

2. Migration Plan
3. DMS Configuration Document
4. Validation Report
5. Cutover Checklist
6. Rollback Plan
7. Migration Completion Report